

PROJECT PLAN

WINNEST

BREEDING INTELLIGENCE PLATFORM



TEAM IT-1A

INFORMATION TECHNOLOGY PROGRAMME

YEAR 1 PERIOD 3

NHL STENDEN HOGESCHOOL

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1. Executive Summary

WINNEST is a breeding intelligence platform designed to support racing pigeon breeders in managing breeding activities and identifying the most successful breeding pairs. The project was developed in response to the limitations of traditional breeding management methods, which often rely on handwritten notes and spreadsheets.

The platform combines breeding records, youngster information, health records, race results, and performance analytics into a centralized system. Using pair ranking and breeding performance analysis, breeders can make more informed breeding decisions.

This project was completed as part of the Project Innovate module and involved research, design, development, testing, and documentation activities.

2. Project Overview

Background

Racing pigeon breeders invest significant time and resources into breeding and racing pigeons. One of the most important questions breeders face each year is determining which breeding pairs produce the best-performing offspring.

Although race administration software such as Compustam exists, these systems primarily focus on race performance and provide limited support for breeding pair analysis.

As a result, many breeders continue to use handwritten notes and personal experience when selecting breeding pairs for future seasons.

Project Description

WINNEST was developed as a breeding intelligence platform that records breeding activities and combines them with race performance data to generate meaningful analytics for breeders.

3. Project Objectives

Main Objective

Develop a web-based breeding intelligence platform that helps pigeon breeders identify their most successful breeding pairs.

Goals

- Digitalize breeding management.
- Reduce dependence on handwritten records.
- Improve breeding decision-making.
- Provide pair performance analytics.
- Create a user-friendly system suitable for breeders.

4. Project Scope

In Scope

- User authentication
- Dashboard
- Pair Management
- Pair Details
- Nest Management
- Youngster Management
- Youngster Profiles
- Health Records
- Race Results
- Analytics Dashboard

Out of Scope

- Mobile application
- AI prediction engine
- GPS tracking
- DNA analysis integration
- Online marketplace

5. Stakeholders

Primary Stakeholders

- Racing pigeon breeders
- Project client

Secondary Stakeholders

- Development team
- NHL Stenden teachers
- Future users

6. Team Structure and Responsibilities

Role	Team Member	Role Description
Project Leader	Caleb Gaitou	Responsible for coordinating the project and ensuring that the team worked effectively toward the project goals.
UX/UI Designer	Jenjira Huaisai	Responsible for designing the user experience and user interface of the WINNEST platform to ensure that the system is easy to use, visually

Role	Team Member	Role Description
		appealing, and suitable for pigeon breeders
Frontend Developer	1. Jenjira Huaisai 2. Oleksii Khomiak	Responsible for implementing the user interface and ensuring that users could interact with the system effectively through a web browser.
Backend Developer	1. Caleb Gaïtou 2. Rodrigo de Araújo Castanheira	Responsible for implementing the business logic and server-side functionality of the system.
Database Engineer	Muhammad Umar Kamran	Responsible for designing and managing the database structure used to store breeding, youngster, health, and race data.
Team Collaboration	All team members	Regularly participate in discussions, provide feedback, solve problems together, and

Role	Team Member	Role Description
		support one another throughout the project lifecycle.

7. Project Methodology

The project followed an Agile-Development Approach.

The team conducted weekly meetings to discuss progress, assign tasks, and review completed work.

Project Milestone

Milestone	Week
Project Idea Approved	Week 1
Initial Poll	Week 2
Functional Design Completed	Week 3
Database Design Completed	Week 4
Intermediate Poll	Week 5
MVP Prototype Completed	Week 6
Documentation Completed	Week 7
Final Poll	Week 8

8. Work Breakdown Structure

Research

- Problem analysis
- User needs analysis
- Competitor research

Design

- Functional design
- Database design
- Wireframes
- High-fidelity UI

Development

- Frontend development
- Backend development
- Database implementation

Testing

- Functional testing
- Bug fixing

Delivery

- Documentation
- User manual
- Final presentation

9. Project Schedule

Week	Activity
1	Research
2	Initial Poll
3	Functional Design
4	Database Design
5	Intermediate Poll
6	Development
7	Testing & Documentation
8	Final Poll

10. Network Planning

Requirements Analysis



Functional Design



Database Design



UX/UI Design



Development



Testing



Documentation



Final Product

11. Gantt Chart

Activities	W1	W2	W3	W4	W5	W6	W7	W8
Project Idea & Problem Identification	■							
User & Market Research	■	■						
Initial Poll Preparation	■	■						
Initial Poll Presentation		■						
Requirements Analysis		■	■					
Functional Design		■	■					
Competitor Analysis (Compustam)		■	■					
Database Design (ERD)			■	■				
UX Wireframes			■	■				
High-Fidelity UI Design			■	■	■			
Intermediate Poll Preparation				■	■			
Intermediate Poll Presentation					■			
Frontend Development				■	■	■		
Backend Development				■	■	■		
Database Implementation				■	■	■		
Testing & Bug Fixing						■	■	
User Manual Creation						■	■	
Technical Advice Report						■	■	■
Project Plan Documentation						■	■	■
Final Poll Preparation							■	■
Final Poll Presentation								■

12. Resource Planning

Software

- Figma
- Visual Studio Code
- phpMyAdmin
- MySQL
- GitHub

Hardware

- Personal computers
- Internet connection

13. Risk Management

Risk	Impact	Mitigation
Scope creep	High	Focus on MVP
Time limitations	High	Weekly planning
Technical challenges	Medium	Team collaboration
Team member absence	Medium	Documentation and communication

14. Project Deliverables

Documentation

- Project Plan

Technical Advice Report

- Functional Design
- User Manual

Technical Deliverables

- Source code
- Database
- GitHub repository

Design Deliverables

- Wireframes
- High-fidelity screens
- Design system

15. Project Monitoring and Control

Progress was monitored through:

- Weekly team meetings
- Task tracking
- Initial, intermediate and final polls feedback
- Teacher consultations

- GitHub version control

16. Project Evaluation

The project successfully delivered a breeding intelligence platform concept that addresses a significant challenge faced by pigeon breeders.

The platform centralizes breeding information and provides analytics that help breeders make informed breeding decisions.

17. Conclusion

WINNEST demonstrates how digital technology can support traditional pigeon breeding management. By combining breeding records with race performance data, the system enables breeders to identify successful breeding pairs and improve future breeding strategies.

18. Appendices

The repository contains all supporting documentation and project deliverables referenced in this report, including:

- Meeting minutes
- Figma designs and prototypes
- Source code

- Presentation materials
- Functional Design document
- Technical Design document
- Database design (ERD)
- Project planning documents
- Technical Advice Report
- Other relevant project documentation

GitHub Repository:

<https://github.com/rodrigo-castanheira/Project-Winnest.git>



WINNEST

Breeding Intelligence

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